

Utilizing Sentiment Insights for Software Evaluation with Aspect-Based Sentiment Analysis in App Reviews Using IndoBERT-Lite

Nikita Ananda Putri Masaling

Computer Science Department,

BINUS Graduate Program - Doctor of Computer Science

Bina Nusantara University

Jakarta, Indonesia

nikita.masaling@binus.ac.id

Yulyani Arifin

Computer Science Department,

BINUS Graduate Program - Doctor of Computer Science

Bina Nusantara University

Jakarta, Indonesia

yulyaniarifin@binus.ac.id

Haryono Soeparno

Computer Science Department,

BINUS Graduate Program - Doctor of Computer Science

Bina Nusantara University

Jakarta, Indonesia

haryono@binus.edu

Ford Lumban Gaol

Computer Science Department,

BINUS Graduate Program - Doctor of Computer Science

Bina Nusantara University

Jakarta, Indonesia

fgaol@binus.edu

Abstract— With mobile applications increasingly used daily, understanding user feedback is vital for software improvement. Manual analysis of user reviews on platforms is time-consuming and lacks scalability. This study utilizes aspect-based sentiment analysis with the IndoBERT-Lite model to automate the categorization and sentiment detection of feedback. Despite advancements, precise aspect classification and sentiment detection in Indonesian language reviews remain challenging due to limited resources. Using 5,000 Tokopedia reviews, labeled across five aspects, namely functionality, service, performance, usability, and design, the model achieved optimal results with batch size 32, showing 0.85 accuracy and 0.866 F1 score for aspect classification, and 0.998 across all metrics for sentiment classification.

Keywords—Aspect-Based Sentiment Analysis, Fine-tuning, IndoBERT-Lite, User Feedback

I. INTRODUCTION

The software development life cycle (SDLC) encompasses a variety of methods and practices aimed at guiding the systematic creation of software applications. These methods can include phases such as planning, analysis, design, implementation, testing, deployment, and maintenance. Regardless of the various SDLC methodologies, one of the stages that is always carried out is evaluation. Evaluation is one of the most important stages in SDLC because by conducting evaluation, developers can ensure that the software meets the requirements and quality standards that have been previously set. Evaluation of software can be done by reviewing code, testing performance, analyzing usability, and collecting user feedback. By evaluating software, developers can find out and avoid any problems or deficiencies before the product is released to the market [1], [2], [3], [4].

In previous practice, software system evaluation was typically conducted during the system testing phase. Over time, the focus of the evaluation phase has shifted from treating evaluation solely as a final checkpoint intended for validation, to embracing it as an ongoing process integral to iterative design. This change has made evaluation not only be

done to measure software performance, but as an important stage for gathering information and feedback that can improve the quality of the product being developed. Thus, software can be developed in a more responsive and user-centric way [5].

One effective way to evaluate software to ensure it meets customer needs is to collect user reviews or opinions. Many mobile app store platforms have review features that allow users to rate or share their opinions about the software they have installed. One of them is the Google Play Store. User reviews can give valuable insights for developers, which can be used as a tool for software evaluation and improvement [6].

Although user reviews present in the review feature of app store platforms offer valuable feedback, they are not enough to provide actionable insights for developers for software evaluation. Additional analysis is needed to extract information that can improve software quality. Sentiment analysis is one method used to gain insight from user opinions expressed in review texts[7]. With the development of research on sentiment analysis, a new technique called Aspect-Based Sentiment Analysis (ABSA) has been developed, a technique that provides more detailed information about sentiments related to specific aspects of a domain in a text dataset [8]. Using ABSA, developers can understand user ratings of certain features, allowing them to identify which aspects need improvement. Figure 1 shows how aspect-based sentiment analysis can show sentiment for a particular aspect.



Fig. 1. Example of aspect-based sentiment analysis

Fine-grained sentiment analysis such as ABSA was initially often performed using traditional machine learning techniques such as lexicon-based, Support Vector Machines, Naive Bayes, K-Nearest Neighbors, etc. Although these methods produce quite good performance, they usually require large datasets and still need to be improved [9]. Along with time, and the development also increasing research that has been conducted about machine learning, more advanced models like deep learning and transformer-based models such as BERT, RoBERTa, BART, etc., have been used for ABSA, and these models have shown significant improvements in accuracy and effectiveness [10], [11], [12], [13]. Therefore, this study proposes the use of transformer-based models to improve accuracy while maintaining model efficiency, especially for aspect-based sentiment analysis tasks.

The transformer-based model that is used in this study is IndoBERT-Lite developed by the IndoLEM team. This IndoBERT-Lite model is a model specifically designed for Indonesian text data. IndoBERT-Lite is optimized with fewer parameters, allowing faster data processing without reducing effectiveness [14], [15]. This model is suitable for the dataset that will be used in this experiment, which is in Indonesian language, because IndoBERT-Lite has adaptability for Indonesian language-specific and it is also proven efficient, making it ideal for handling large-scale Indonesian text data in aspect-based sentiment analysis. The selection of IndoBERT-Lite allows a balance between accuracy and computational speed, which is very important for practical applications.

II. RELATED WORKS

A. Aspect-Based Sentiment Analysis (ABSA)

Aspect-Based Sentiment Analysis (ABSA) is a more complex task compared to regular sentiment analysis that only classifies text into certain polarities. Instead of simply determining the overall sentiment of a text, ABSA determines certain attributes or aspects of the identified text and determines the sentiment polarity classification of the identified text. This approach was introduced at the SemEval-2014 competition. At this event, text dataset with annotated reviews with the domain aspects of restaurants and laptops was also proposed [16].

Early research on Aspect-Based Sentiment Analysis (ABSA) often uses traditional machine learning models, like lexicon-based method, Support Vector Machines (SVM), Naive Bayes, K-Nearest Neighbors (KNN), and more [17], [18]. One of them used SVM for ABSA and achieved an average accuracy rate of 77.98% [19]. When it comes to classifying sentiment polarity, these early methods were useful, but they needed more time for feature engineering and pre-processing, and they weren't able to handle complex sentiment expressions in text.

As the field advanced, more sophisticated approaches emerged, with deep learning and transformer-based models becoming the standard due to their ability to handle complex patterns and contextual information in text [8], [20]. Despite these advancements, research on ABSA for low-resource languages, such as Indonesian, remains underdeveloped. This gap highlights the need for further studies and model

adaptation to improve sentiment analysis capabilities in languages with limited annotated data and resources.

B. IndoBERT-Lite Model

IndoBERT-Lite is a one of the IndoNLU model [15]. It is based on the ALBERT architecture, which is an improvement on BERT that makes computations easier by using parameter factorisation and weight sharing [21]. The large Indo4B dataset, which has about 4 billion words from 250 million Indonesian lines, was used to train this model design. IndoBERT-Lite has gained good performance in tasks like sequence labelling and classification, therefore, it can be used for a variety of natural language processing tasks in Indonesian. The improved IndoBERT-Lite-SQuAD model stands out because it achieves a peak accuracy of 80% and an F-score of 89%, making it a better model in this design for certain NLP tasks [22]. This success shows how well it works for task that need to understand complex words while still being quick.

III. METHODOLOGY

This section presents the methodology employed to address the challenge of aspect-based sentiment analysis in evaluating mobile applications using user feedback. It covers the research flow, including dataset collection and preparation, the configuration of hyperparameters for the experiments, and the evaluation procedures implemented.

A. Experiment Flow

As part of this experiment, a number of procedures were carried out in order to attain a high level of accuracy in aspect-based sentiment analysis, which ultimately made it possible to conduct user-centred software evaluations that were effective. Starting with the scraping of data using specified parameters that were in line with the assessment goals was the first step in the procedure. Following the completion of the data gathering process, the reviews were subjected to human labelling process in order to classify them into five aspect categories and three sentiment polarity categories. Following the completion of preprocessing, the data might then be utilised for the purpose of training the model. In order to do this, the data has to be compartmentalised into three subsets: training, validation, and testing. After that, tokenisation was performed by utilising the tokeniser that was already included into the IndoBERT model.

With the dataset prepared, the next step was to set up the model itself. Hyperparameter tuning was conducted to optimize the model's performance for the specific dataset. Subsequently, the IndoBERT-Lite model was trained using the prepared data. Finally, an evaluation was performed to assess whether the trained model exhibited sufficient performance for the task of aspect-based sentiment analysis on the mobile app reviews dataset. Figure 2 shows the experiment flow of this research.

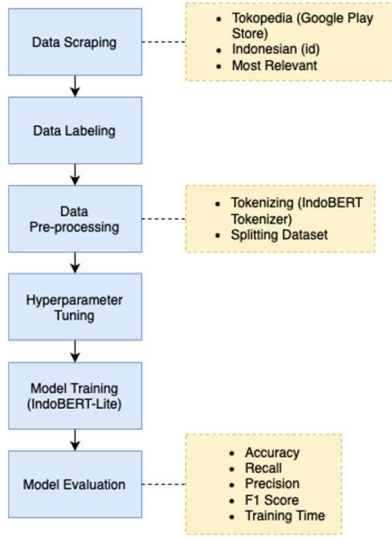


Fig. 2. Experiment flow

B. Dataset

The dataset for this experiment was obtained by scraping the Google Play Store using the Google-Play-Scraper library. This tool enables more specific collection of application reviews by allowing selection based on user language and country. Additionally, it permits sorting of reviews, in this case, reviews were sorted by relevance to so the quality of the sentiments that are scraped will be better. The number of reviews to be scraped is also customizable, and for this experiment, 5,000 user reviews were collected. The dataset was scraped from user reviews of the Tokopedia application, an Indonesia-based e-commerce platform. Table 1 shows the parameters used in scraping the dataset.

TABLE I. DATASET PARAMETERS

Parameter	Value
Application ID	com.tokopedia.tkp
Language	Indonesian (id)
Country	Indonesian (id)
Sorting Criteria	Most Relevant
Number of Reviews	5,000

After gathering the dataset through the scraping process, the next step is to label the collected data. This labeling was conducted using manual annotation, allowing for precise control over the accuracy of the labels. The review text is categorized into five aspect categories: design, functionality, performance, service, and usability. Each review text can have one or more of these aspects, recognizing that user feedback often touches on multiple facets of the application. It is also possible to divide the polarity of sentiment into three distinct categories: negative, neutral, and positive. By utilising this dual labelling of aspects and polarity of sentiment, it is possible to conduct a comprehensive study of user views, which in turn enables a more targeted approach to the assessment and improvement of software. Detailed information on the composition of the dataset is included in

Table 2, which displays the statistical distribution of each label that has been annotated.

TABLE II. LABEL DISTRIBUTION

	Negative	Neutral	Positive
Design	36	18	77
Functionality	646	646	1094
Performance	567	52	175
Service	957	321	833
Usability	304	86	370

After the dataset was finished being labelled, it was then separated into three independent subsets: the training dataset, the validation dataset, and the testing dataset. The dataset was divided into three distinct parts: 70% for training, 20% for validation, and 10% for testing. Using this proportionate divide, not only does the model have access to a substantial amount of data from which it may learn, but it also provides independent datasets for the purpose of adjusting and assessing its performance. Following that, the complete dataset was subjected to tokenization, which was carried out with the help of the IndoBERT model's built-in tokenizer. This process includes turning words into numerical representations in order to get the text data ready for introduction into the model.

C. Hyperparameter Setting

The batch size is the primary focus of the hyperparameter settings that are being observed in this experiment. The numbers 16, 32, and 64 of batch size are being evaluated. As a result of the fact that it can have a substantial influence on both performance and the effectiveness of training, the selection of batch size is an essential component of model tuning [23], [24]. The goal of our experiments is to discover the ideal value that will deliver greater model performance while preserving optimal training duration. This will be accomplished by testing with different batch sizes. This meticulous tuning contributes to the achievement of a balance between accuracy and computing resources, which eventually results in an improvement in the efficiency of aspect-based sentiment analysis.

D. Evaluation

The performance of the trained model is evaluated using several metrics. These metrics are training time, accuracy, precision, recall, and F1 score. These measures give a whole knowledge of the possibilities of the model by assessing several facets of its performance in aspect-based sentiment analysis.

By finding how often the model correctly predicts the sentiment labels across all categories, accuracy evaluates its general performance. By computing the ratio of accurately predicted results to the whole number of predictions generated, it offers a broad estimate of the model's performance. Precision measures how well the model distinguishes among all the positive events it predicts as such. It indicates the proportion of the expected positive outcomes that are truly positive, therefore emphasizing the quality of positive forecasts. High accuracy indicates that the model

generates low false positive rate. True Positives (TP) in this context are accurately expected positive sentiment. False Positives (FP) are situations when the model mistakenly forecasts a positive category for a negative sentiment. Recall evaluates the model's capacity to find every real positive sentiment in the dataset. It shows how many of the real positive instances the model effectively gathers, therefore reflecting the completeness of the positive predictions of the model. Excellent recollection points to a low false negative rate.

In addition to measuring model performance, the speed in training is also measured. Training time measures the efficiency of the model in terms of the time taken to learn from the training data. It is an important practical consideration, especially when working with large datasets or when computational resources are limited. Shorter training times make the model more scalable and cost-effective. Training time is typically recorded in seconds, minutes, or hours, measured from the start to the completion of the training process.

IV. RESULTS AND DISCUSSION

The experimental results show the performance of the IndoBERT-Lite model, evaluated using various metrics discussed in the previous section. Among the hyperparameter configurations tested, a batch size of 32 got the best performance across most metrics, suggesting that this batch size optimally balances model accuracy, precision, recall, and F1 score. Table 3 compares the performance of the IndoBERT-Lite Base model across different batch sizes (16, 32, and 64) based on several evaluation metrics in both the Aspect and Sentiment categories.

TABLE III. PERFORMANCE OF MODEL

Aspect				
Batch Size	Accuracy	Recall	Precision	F1 Score
16	0.840	0.840	0.818	0.829
32	0.850	0.850	0.883	0.866
64	0.79	0.79	0.844	0.816
Sentiment				
Batch Size	Accuracy	Recall	Precision	F1 Score
16	0.948	0.948	0.956	0.952
32	0.998	0.998	0.998	0.998
64	0.974	0.974	0.975	0.974

Using an NVIDIA T4 GPU, the experiment shows that the training time for a batch size of 32 is slightly longer, 37 seconds more than the model trained with a batch size of 64, but the gains in accuracy and consistency make this configuration worthwhile. The minor increase in training time is thus considered a reasonable trade-off for the enhanced model performance in aspect-based sentiment analysis. Table 4 shows the training time for each batch size settings.

TABLE IV. TRAINING TIME OF EACH SETTINGS

Batch Size	Training Time
16	0:15:17
32	0:12:51
64	0:12:14

Figure 3 shows the training and validation loss for both aspect-based classification (left) and sentiment classification (right) over three epochs. In both graphs, there is a noticeable downward trend, indicating that the model's loss decreased as training progressed. For aspect-based classification, the training loss begins above 0.6 and declines sharply, with the validation loss following a similar pattern but with a consistently lower value. The sentiment classification loss exhibits a similar trend, starting with a relatively high loss that decreases significantly across epochs. This reduction in loss demonstrates effective learning, with the model showing improvements in both tasks as it is trained.

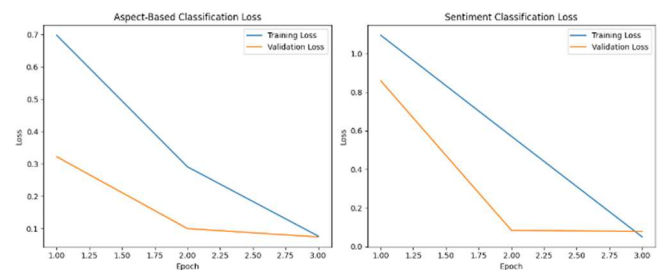


Fig. 3. Training loss

Figure 4 shows confusion matrices for aspect-based and sentiment classification, showing the model's accuracy in predicting different categories. In the aspect-based classification matrix (left), the highest accuracy is observed in the functionality and service aspects, with many correct predictions (e.g., 189 for functionality and 210 for service) and fewer misclassifications across other categories. The sentiment classification matrix (right) shows a strong ability to correctly predict the positive and negative sentiment labels, with 204 correct positive predictions and 194 correct negative predictions. The model has a few misclassifications in the neutral category, indicating that while the model effectively captures positive and negative sentiments, neutral sentiments are occasionally less clear.

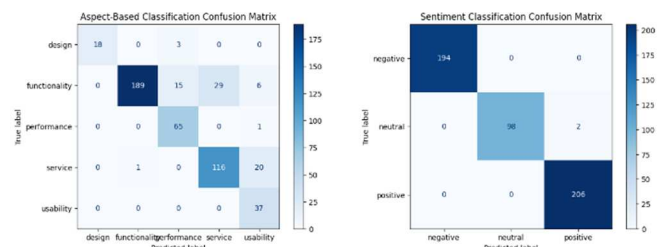


Fig. 4. Confusion matrix

Figure 5 shows the comparison of sentiment distribution across each aspect category in a stacked bar chart. This chart shows that functionality and service aspects have the highest count of user sentiments, with a notable mix of positive, neutral, and negative feedback. Performance, usability, and

design aspects have lower counts. This distribution provides insights into user satisfaction across different aspects, indicating areas that may require more attention, such as design, which received the least positive feedback.

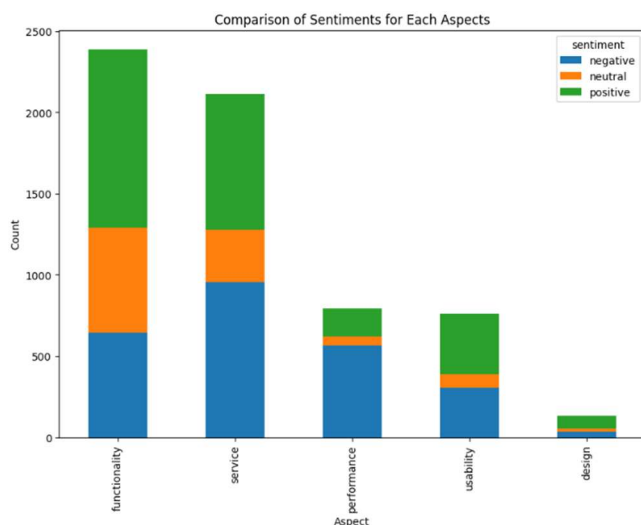


Fig. 5. Sentiment distribution across each aspect category

V. CONCLUSION

This research highlights the successful application of aspect-based sentiment analysis on mobile application reviews, focusing on user feedback to identify key areas for software improvement. Using IndoBERT-Lite, the model achieved high performance across multiple evaluation metrics with an optimal batch size setting of 32. For aspect classification, the model achieved an accuracy of 0.850, recall of 0.850, precision of 0.883, and an F1 score of 0.866. In sentiment classification, it reached even better performance, with accuracy, recall, precision, and F1 score all at 0.998. The analysis of user feedback further revealed that the functionality and service aspects received the most attention, with functionality showing a higher proportion of feedback, suggesting a potential area for development focus. These insights allow for a targeted approach in software evaluation, guiding developers to prioritize specific improvements aligned with user expectations.

Future research could aim to refine the model's ability to capture neutral sentiments, as this category presented some challenges. Additionally, expanding the dataset to include reviews from other applications and domains would help generalize the findings and validate the model's robustness. Then, the performance of IndoBERT-Lite could be compared with other pre-trained language models for Indonesian. Exploring other machine learning techniques or incorporating more advanced language models could also improve classification accuracy and broaden the applicability of aspect-based sentiment analysis in software evaluation.

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